

Lecture 25: Fairness

CS 580B1: Trustworthy Machine Learning

Fall 2024

Acknowledgments

Course slides derived from material created by Rajeev Alur and Osbert Bastani at UPenn ([link](#) to their course)

[Fairness and machine learning](#) book by Barocas, Hardt, and Narayanan

[Slides](#) by Justin Johnson and David Fouhey at University of Michigan

[Slides](#) by Roger Grosse et al. at University of Toronto

[Slides](#) by David Madras at University of Toronto

Other relevant courses on Trustworthy Machine Learning:

[CS 329T](#) at Stanford

[CS 521](#) at UIUC

[ECE1784H/CSC2559H](#) at U Toronto

Course logistics

- Reviews 11 and 12 only for extra credit (10%)

Bias in machine learning

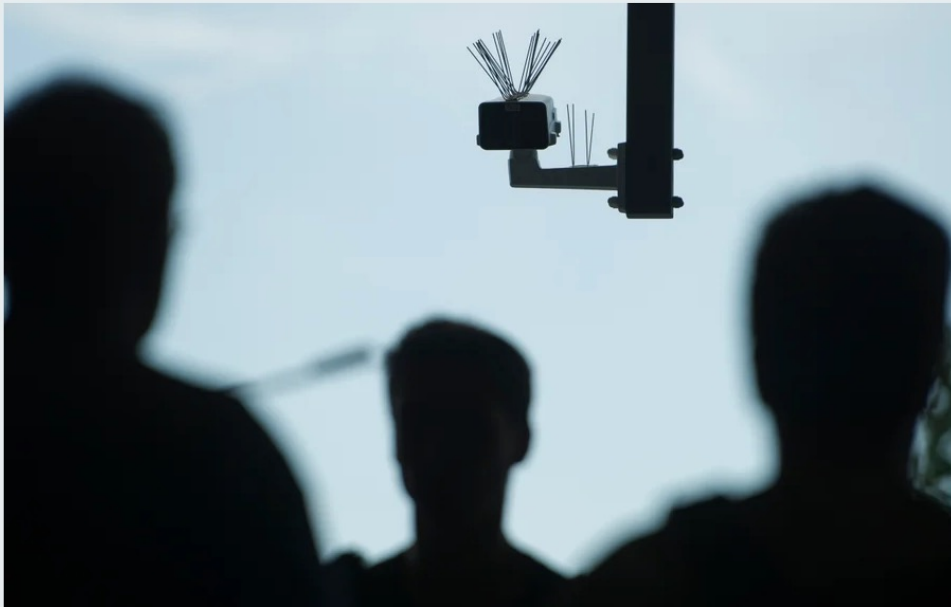
OPINION

MAY 18, 2023 | 5 MIN READ

Police Facial Recognition Technology Can't Tell Black People Apart

AI-powered facial recognition will lead to increased racial profiling

BY THADDEUS L. JOHNSON & NATASHA N. JOHNSON



Steffi Loos/Getty Images

Science

Current Issue First release papers Archive A

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RESEARCH ARTICLE



Dissecting racial bias in an algorithm used to manage the health of populations

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129,494 1,266



Racial bias in health algorithms

The U.S. health care system uses commercial algorithms to guide health decisions. Obermeyer *et al.* find evidence of racial bias in one widely used algorithm, such that Black patients assigned the same level of risk by the algorithm are sicker than White patients (see the Perspective by Benjamin). The authors estimated that this racial bias reduces the number of Black patients identified for extra care by more than half. Bias occurs because the algorithm uses health costs as a proxy for health needs. Less money is spent on Black patients who have the same level of need, and the algorithm thus falsely concludes that Black patients are healthier than equally sick White patients. Reformulating the algorithm so that it no longer uses costs as a proxy for needs eliminates the racial bias in predicting who needs extra care.

Science, this issue p. 447; see also p. 421

Bias in machine learning

Input: Low-Resolution Face



Output: High-Resolution Face



Bias in machine learning

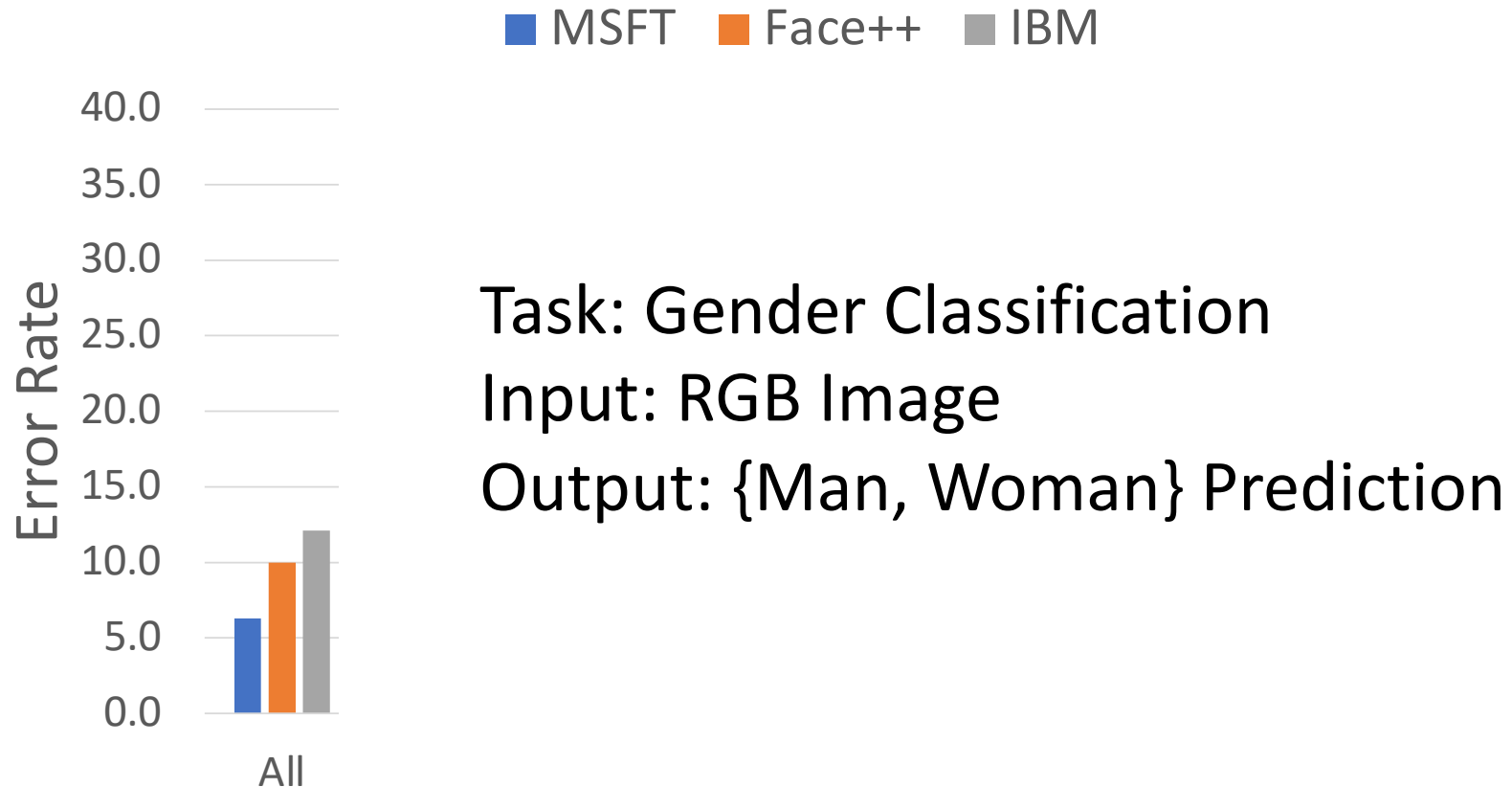
Gender stereotype *she-he* analogies.

sewing-carpentry	register-nurse-physician	housewife-shopkeeper
nurse-surgeon	interior designer-architect	softball-baseball
blond-burly	feminism-conservatism	cosmetics-pharmaceuticals
giggle-chuckle	vocalist-guitarist	petite-lanky
sassy-snappy	diva-superstar	charming-affable
volleyball-football	cupcakes-pizzas	hairstylist-barber

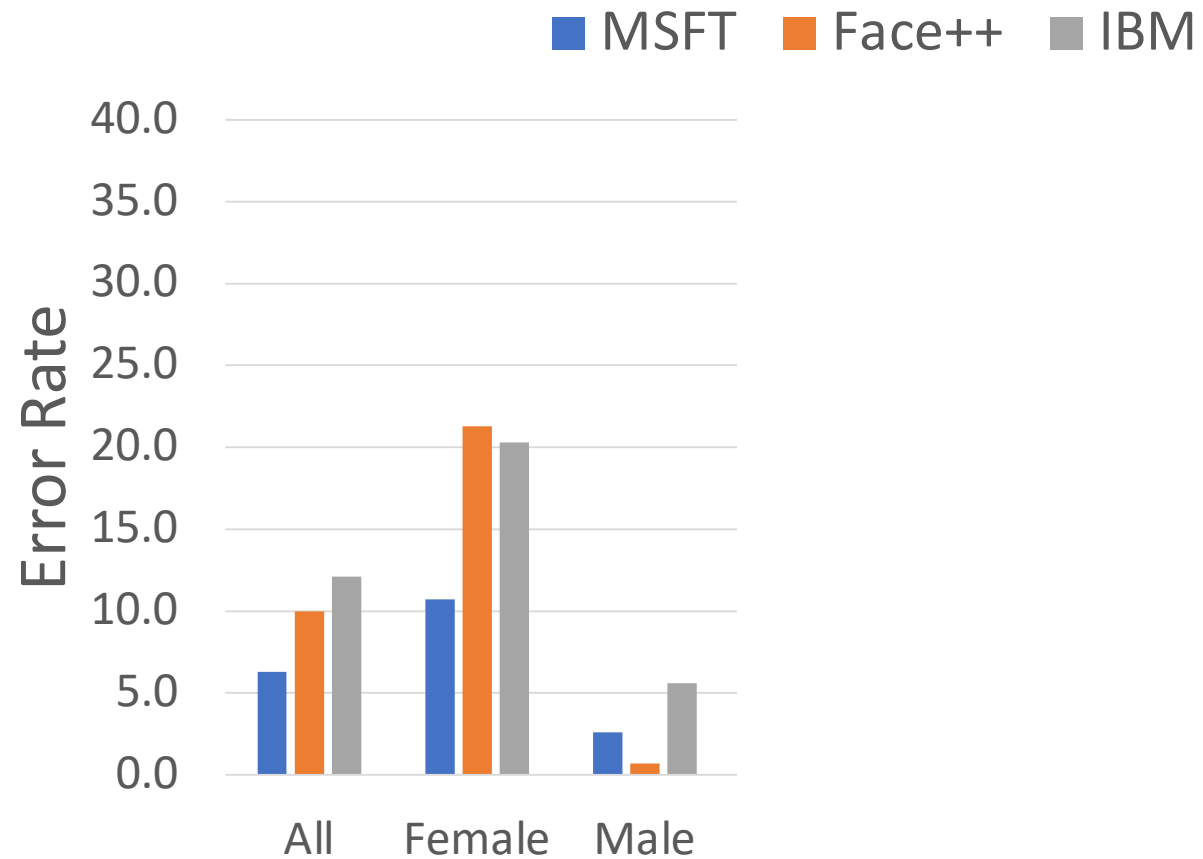
Gender appropriate *she-he* analogies.

queen-king	sister-brother	mother-father
waitress-waiter	ovarian cancer-prostate cancer	convent-monastery

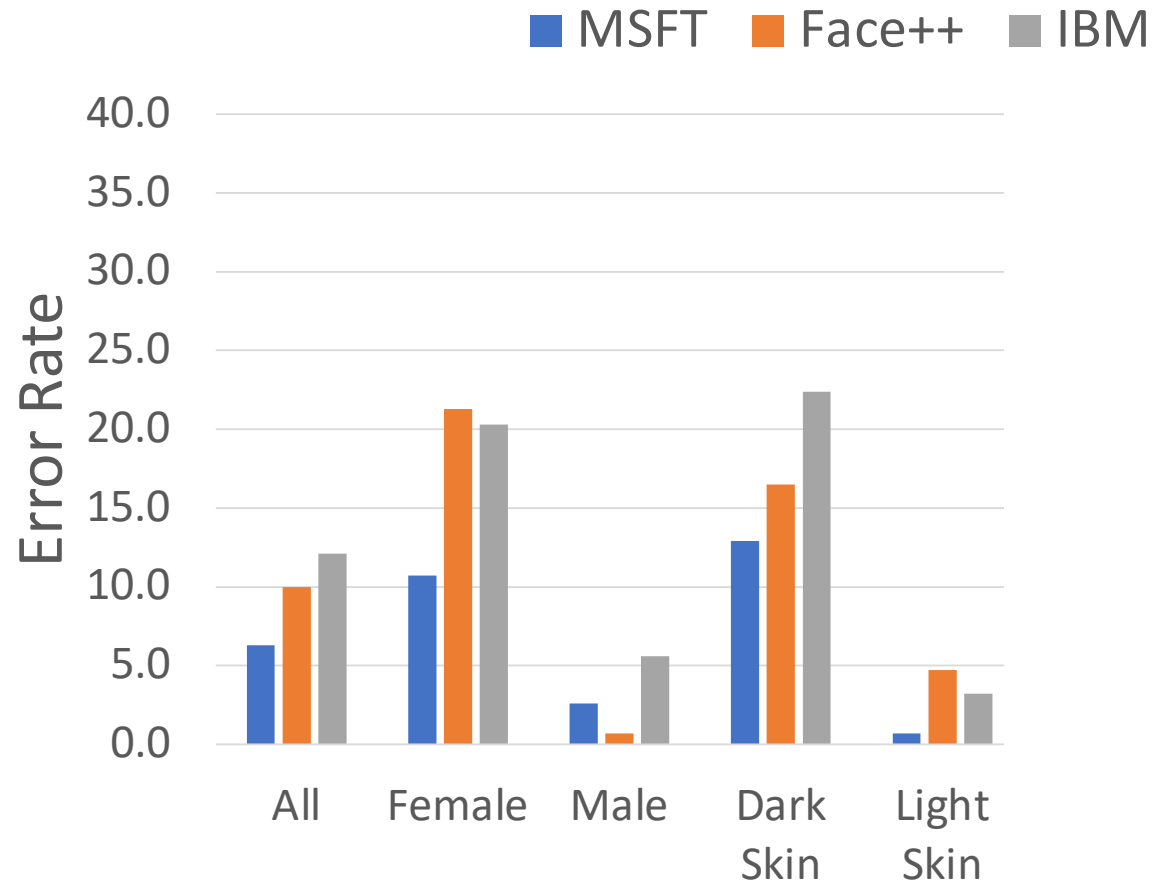
Bias in machine learning



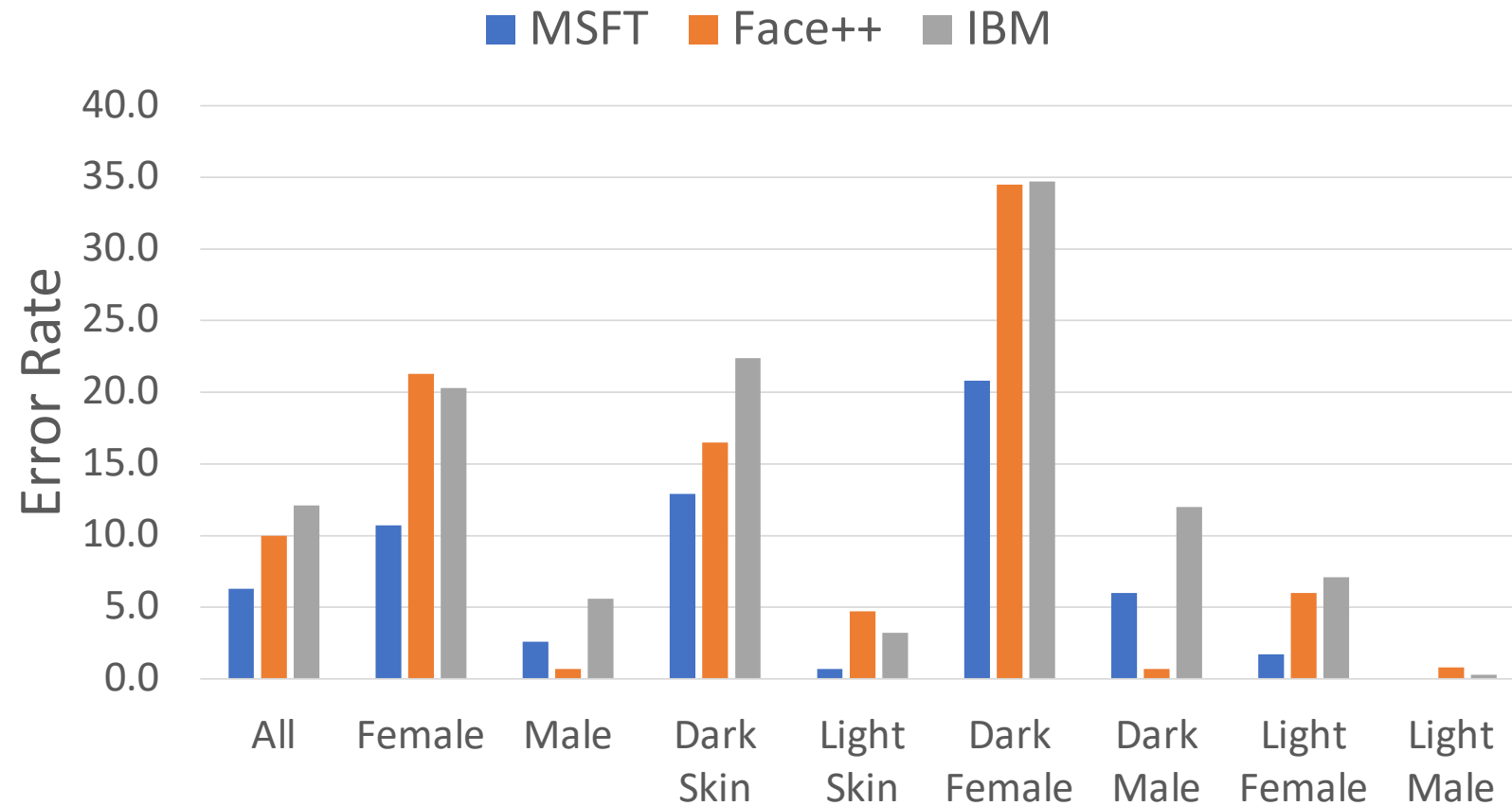
Bias in machine learning



Bias in machine learning



Bias in machine learning



Agenda

- *Sources of bias*
- Fairness definitions: Naïve notions (Fairness through unawareness), Individual fairness
- Fairness definitions: Group fairness
- Other fairness definitions

Sources of bias

- **(Data)** Imbalanced/impoverished data
- **(Data)** Labeled data imbalance
- **(Data)** Labeled data incorrect / noisy (historical bias)
- **(Model)** Imbalanced ML prediction error

Imbalanced/impoverished data

- Less data from minority groups → Higher error on minority groups
- **Example:** Many clinical trials historically recruited largely white males, leading to biases in understanding outcomes and side effects
- **Example:** Focus on easily accessible data (e.g. recent tweets, or easily measured features of people) can lead to biased datasets
- Need to be careful to gather representative datasets

Labeled data imbalance

- Labels *collected* in a biased manner
- **Example:** Predictive policing
 - “PredPol” predictive policing system employed in some police departments
 - Suppose that crime happens equally everywhere
 - Some areas more policed
 - More crime found in those areas
 - ML learns to predict crime in neighborhoods that were more policed

Labeled data incorrect / noisy (historical bias)

- Labels *assigned* in a biased manner
- **Example:** [Amazon hiring bias](#)
 - Amazon's ML resume screening tool to predict hiring decisions based on 10 years of historical applicant data; but found it was biased against women
 - Labels tainted by historical bias

Imbalanced ML prediction error

- Low accuracy of model on minority groups can result in unfairness
- **Example:** Strong student as likely as weak student to be admitted into college for group A but not for group B

Agenda

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Blind fairness (or Fairness through unawareness)

- Predictive model should ignore sensitive attributes
- **Problem:** Other attributes may be correlated with sensitive attributes
 - Race is correlated with poverty
- **Problem:** It is “fair” to randomly predict for one subgroup as long as sensitive attributes are omitted

Blind fairness (or Fairness through unawareness)

- **Legally Protected Attributes**

- Race, sex, color, religion, national origin (Civil Rights Act of 1964, Equal Pay Act of 1963)
- Age (Discrimination in Employment Act of 1967)
- Citizenship (Immigration Reform and Control Act)
- Pregnancy (Pregnancy Discrimination Act)
- Familial status (Civil Rights Act of 1968)
- Disability (Rehabilitation Act of 1973; Americans with Disabilities Act of 1990)
- Veteran status (Vietnam Era Veterans' Readjustment Assistance Act of 1974; Uniformed Services Employment and Reemployment Rights Act)
- Genetic information (Genetic Information Nondiscrimination Act)

Case Study: COMPAS

- Software by Northpointe to predict **recidivism** for defendants
 - i.e., risk of committing future crimes
- Used to help make bail, sentencing, and parole decisions

Table 1: ProPublica Analysis of COMPAS Algorithm

	White	Black
Wrongly Labeled High-Risk	23.5%	44.9%
Wrongly Labeled Low-Risk	47.7%	28.0%

Case Study: COMPAS

- **Features:** 137 questions answered by defendants or criminal records:
 - “Was one of your parents ever sent to jail or prison?”
 - “How many of your friends/acquaintances are taking drugs illegally?”
 - “How often did you get in fights while at school?”
 - Agree or disagree? “A hungry person has a right to steal”
 - Agree or disagree? “If people make me angry or lose my temper, I can be dangerous.”
- Exact algorithm and model is a trade secret

Case Study: COMPAS

- Race is **not** a feature
- Problem: Correlated features
 - E.g., poverty, joblessness and social marginalization
 - One of the developers of the system said it is difficult to construct a score that doesn't include items that can be correlated with race
 - "If those are omitted from your risk assessment, accuracy goes down"

Individual fairness

- “Similar” individuals (differing only on sensitive attributes) should receive “similar” outcomes
- The prediction function $f: X \rightarrow Y$ should be Lipschitz continuous:
$$\|x - x'\| \leq \epsilon \Rightarrow |f(x) - f(x')| \leq \epsilon'$$
- **Problem:** How to define “similar”?
 - What if we include someone’s accent or attire as a feature?
 - Accent may be correlated with race, in which case $\|x - x'\|$ is always large for two individuals of different race, even if race is not included as a feature
- **Problem:** Scales poorly to high-dimensional spaces

Agenda

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- *Fairness definitions: Group fairness*
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Group fairness

- Equalize “fairness metrics” across “subgroups”
- Challenges:
 - Need to define “subgroups” (e.g., ethnicity, gender, etc.)
 - Need to define “fairness metrics” (e.g., rate of positive outcomes, false positive/negative rates, etc.)

Group fairness

- **Problem setup:**

- Sensitive attribute/feature A
- ML model f mapping input features X to prediction $\hat{Y} = f(X)$
- True outcome Y
 - for simplicity, we will assume a binary setting, and $Y = 1$ is the “good” outcome

- **Group fairness:** Account for performance on subgroups

- Require $F(f, A = a) = F(f, A = b)$ where F is a fairness metric

- **Example:**

- Hiring decisions
 - A = gender, $\hat{Y}$ = predicted effectiveness of candidate, Y = true effectiveness of candidate
- Health risk
 - A = race, $\hat{Y}$ = predicted risk of patient, Y = true risk of patient

Group fairness: Independence

- Predictions are independent of sensitive attribute value

$$\Pr(\hat{Y}|A) = \Pr(\hat{Y})$$

$$\text{i.e. } \Pr(\hat{Y} = 1|A = a) = \Pr(\hat{Y} = 1|A = b)$$

- **Example:** Fraction of patients deemed suitable for hiring should be independent of gender
- Also referred to as *demographic parity*, *statistical parity*, *disparate impact*
- **Intuition:** Subgroups should receive positive outcomes at similar rates

Group fairness: Independence

- **Problem:** Can lead to discrimination within the subgroup
 - **Example:** Hire candidates at the same rate in groups a and b but random hiring in group b
 - Not a problem when incentives are aligned (i.e., decision-maker wants to choose best candidates within each subgroup)
 - But this is often not the case! Consider *glass cliffs*
- **Problem:** What if the base rates are not equal?
 - In our health risk example, what if members from a particular subgroup are indeed at a lower risk?

Group fairness: Separability

Event	Condition	Resulting notion ($\mathbb{P}\{\text{event} \mid \text{condition}\}$)
$\hat{Y} = 1$	$Y = 1$	True positive rate, recall
$\hat{Y} = 0$	$Y = 1$	False negative rate
$\hat{Y} = 1$	$Y = 0$	False positive rate
$\hat{Y} = 0$	$Y = 0$	True negative rate

Group fairness: Separability

- Predictions should be independent of sensitive attribute value given ground-truth

$$Pr(\hat{Y}|A, Y) = Pr(\hat{Y}|Y)$$

$$\text{i.e. } Pr(\hat{Y} = 1|Y = 1, A = a) = Pr(\hat{Y} = 1|Y = 1, A = b)$$

$$Pr(\hat{Y} = 1|Y = 0, A = a) = Pr(\hat{Y} = 1|Y = 0, A = b)$$

- False negative rates and false positive rates are equal across subgroups
 - Also referred to as *equality of odds*
-
- **Intuition:** Error rate is equal across subgroups

Case Study: COMPAS

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 - i.e., risk of committing future crimes
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- Violates separability!

Group fairness: Separability

- **Problem:** Can lead to a suboptimal classifier
- **Example:**
 - Consider *equality of opportunity*
 - Similar to separation, but only require that false negative rates are equalized
 - **Intuition:** Only care about being fair to qualified individual
 - Say there are 10 members of group a with $Y=1$ and $\hat{Y}=1$ for all 10; similarly, say there are 10 members of group b but $Y=1$ just for one member and $\hat{Y}=0$ for all 10
 - Accuracy of classifier = $19/20 = 0.95$; FNR for a is 0% but is 100% for b
 - Say, we modify classifier so that $\hat{Y}=1$ for all members of b
 - Now, the classifier satisfies *equality of opportunity* but has accuracy $11/20=0.55$

Group fairness: Sufficiency

- Ground-truth should be independent of sensitive attribute value given prediction

$$Pr(Y|A, \hat{Y}) = Pr(Y|\hat{Y})$$

$$\text{i.e. } Pr(Y = 1|\hat{Y} = 1, A = a) = Pr(Y = 1|\hat{Y} = 1, A = b)$$

$$Pr(Y = 0|\hat{Y} = 0, A = a) = Pr(Y = 0|\hat{Y} = 0, A = b)$$

- Positive and negative predictive values are equal across subgroups

Event	Condition	Resulting notion ($\mathbb{P}\{\text{event} \mid \text{condition}\}$)
$Y = 1$	$\hat{Y} = 1$	Positive predictive value, precision
$Y = 0$	$\hat{Y} = 0$	Negative predictive value

Group fairness: Sufficiency

- Say classifier f outputs probability R instead of binary output $\hat{Y}$
- Recall that f is calibrated if
$$\forall r \in [0,1]. Pr(Y = 1 | R = r) = r$$
- f is *calibrated by group* if for all groups a
$$\forall r \in [0,1]. Pr(Y = 1 | R = r, A = a) = r$$
- **Thm:** *Calibration by group* implies *sufficiency* where *sufficiency* is
$$\forall r \in [0,1]. Pr(Y = 1 | R = r, A = a) = Pr(\hat{Y} = 1 | R = r, A = b)$$

How to satisfy group fairness?

- **Pre-processing:** Adjust the feature space to be uncorrelated with the sensitive attribute.
- **In-training:** Work the constraint into the optimization process that constructs a classifier from training data.
- **Post-processing:** Adjust a learned classifier so as to be uncorrelated with the sensitive attribute.
- *Independence, separability, and sufficiency* are incompatible!
 - Need to carefully choose what notion of fairness to enforce

Related notions of group fairness

List of statistical non-discrimination criteria			
Name	Criterion	Note	Reference
Independence	Indep.	Equiv.	Calders et al. (2009)
Group fairness	Indep.	Equiv.	
Demographic parity	Indep.	Equiv.	
Conditional statistical parity	Indep.	Relax.	Corbett-Davies et al. (2017)
Darlington criterion (4)	Indep.	Relax.	Darlington (1971)
Equal opportunity	Separ.	Relax.	Hardt, Price, Srebro (2016)
Equalized odds	Separ.	Equiv.	Hardt, Price, Srebro (2016)
Conditional procedure accuracy	Separ.	Equiv.	Berk et al. (2017)
Avoiding disparate mistreatment	Separ.	Equiv.	Zafar et al. (2017)
Balance for the negative class	Separ.	Relax.	Kleinberg et al. (2016)
Balance for the positive class	Separ.	Relax.	Kleinberg et al. (2016)
Predictive equality	Separ.	Relax.	Corbett-Davies et al. (2017)
Equalized correlations	Separ.	Relax.	Woodworth (2017)
Darlington criterion (3)	Separ.	Relax.	Darlington (1971)
Cleary model	Suff.	Relax.	Cleary (1966)
Conditional use accuracy	Suff.	Equiv.	Berk et al. (2017)
Predictive parity	Suff.	Relax.	Chouldechova (2016)
Calibration within groups	Suff.	Equiv.	Chouldechova (2016)
Darlington criterion (1), (2)	Suff.	Relax.	Darlington (1971)

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- Sources of bias
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- *Other fairness definitions*

Other fairness definitions

- Causal fairness
 - Kusner et al., “Counterfactual Fairness”, NeurIPS’17
- Multicalibration
 - Hebert-Johnson et al., “Multicalibration: Calibration for the (Computationally-Identifiable) Masses”, ICML’18
- ...

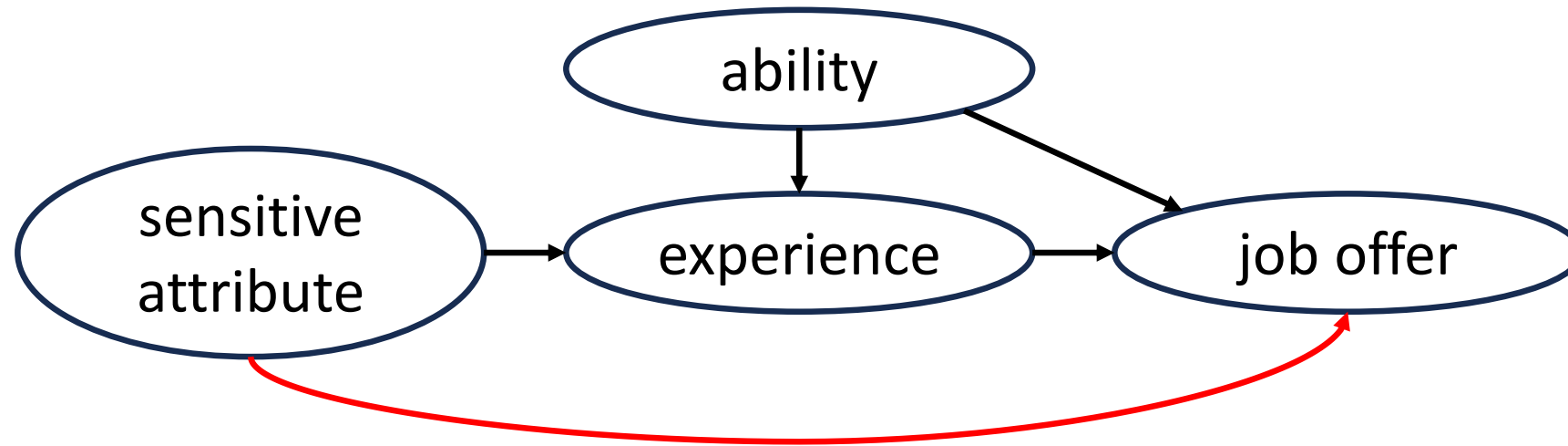
Causal fairness

- Fairness definitions discussed so far ignore the difference between causation and correlation
- **Intuitive definition:** The individual should be treated the same as if you changed their sensitive attribute
- **Example:**
 - If this applicant's sensitive attribute is changed, would you give them the job?
 - If so, then you should give them the job!

Causal fairness

- **Challenge:** How to formalize “changed their sensitive attribute”?
 - Maybe if their sensitive attribute were different, they would have had better job experience by avoiding past discrimination
 - How should this be counted?
- Formalized via counterfactual reasoning

Causal fairness



Takes into account effect of sensitive attribute on experience when changing sensitive attribute

Multicalibration

- In group fairness, if the subgroups are disjoint, then post-hoc enforcing fairness is typically easy
 - In calibration, we simply calibrate each subgroup separately
- What if the subgroups overlap?
 - E.g., Gender, ethnicity, age
 - We could calibrate for each intersection of subgroups, but there are exponentially many subgroups in the number of sensitive attributes
- Multicalibration presents algorithms to ensure calibration for all subgroups in a computationally and statistically efficient way